Strategic Financial Analysis: Freight Cost Prediction & Invoice Risk Flagging

Objective: Minimizing Financial Leakage through Automated Prediction and Anomaly Detection.

1. Executive Summary

his report details the end-to-end development of two critical machine learning pipelines designed to minimize financial leakage: a **Freight Cost Estimator** and an **Automated Invoice Risk Classifier**. Historically, manual cost estimations and broad-stroke invoice approvals have left the organization vulnerable to carrier overcharging, duplicate billing, and audit fatigue.

By transitioning to predictive analytics, this project establishes highly actionable, data-driven financial controls.

Freight Cost Prediction: A Linear Regression model successfully predicts shipping costs with **96.99% accuracy** (an average error of just $24.11), establishing a reliable mathematical baseline for setting strict budget caps.

Invoice Risk Flagging: An optimized Random Forest anomaly detection model identifies statistically suspicious invoices with **96% precision**. This ensures the finance team focuses human labor only on verified risks while confidently automating safe transactions.

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2. Freight Cost Prediction (The Regressor)

2.1 Model Benchmarking & Selection

To ensure a rigorous evaluation, three algorithms were benchmarked: Linear Regression, Decision Trees, and a Random Forest Regressor. Counterintuitively, the simplest model—**Linear Regression**—outperformed the complex ensemble methods.

By prioritizing Occam's Razor, we deployed the Linear Regression model as the final solution. It offers both predictive superiority and high interpretability for the finance team (avoiding the "black box" problem of complex algorithms).

Model	R^2 Score	Mean Absolute Error (MAE)	Root Mean Squared Error (RMSE)	Selection Status
Linear Regression	**96.99%**	**$24.11**	**124.72**	**Deployed (Winner)**
Random Forest Regressor	96.64%	$27.20	131.76	Candidate
Decision Tree Regression	96.30%	$38.12	138.25	Candidate

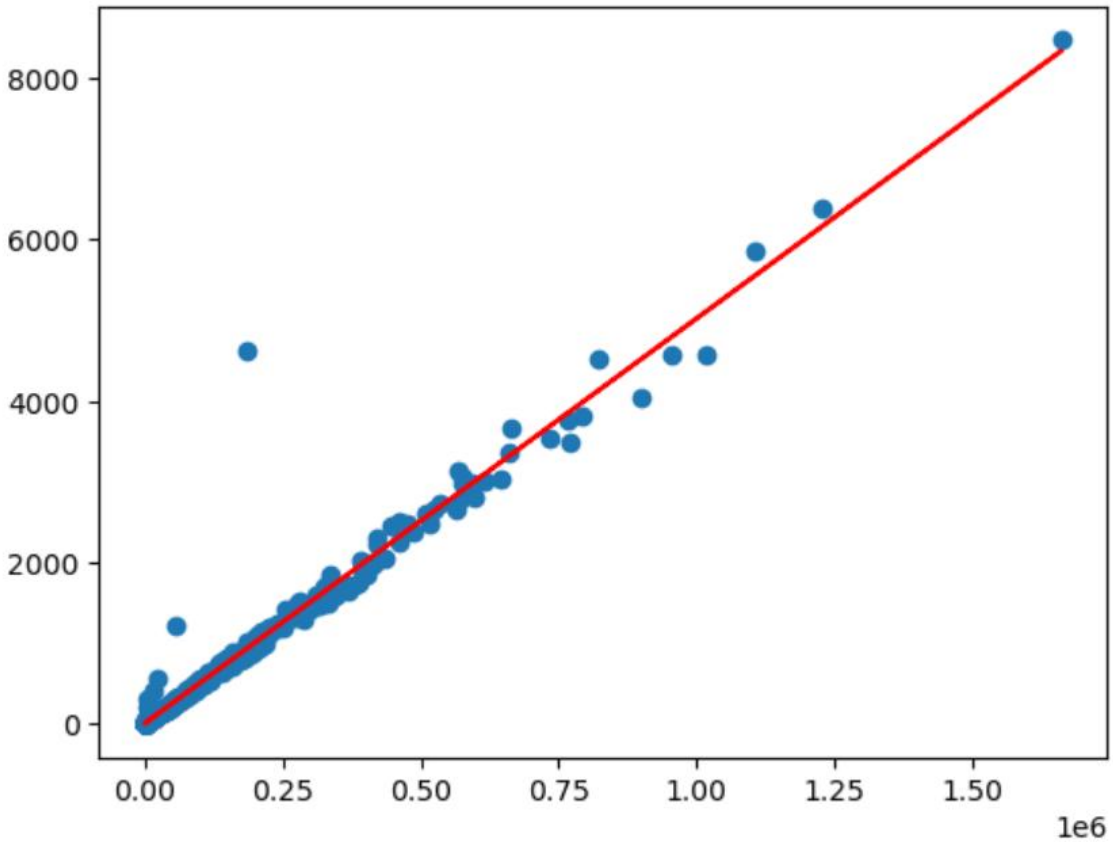

Figure 1: Scatter plot between the predicted (red) and original (blue)

2.2 Business Impact of Cost Prediction

- **96.99% Variance Captured:** We have eliminated almost all "guesswork." The model captures nearly 97% of the factors that influence shipping price.
- **High-Precision Budgeting:** Because the model is accurate to within **$24.11** on average (MAE), the finance team can now set 'Hard Caps' on shipping budgets with extreme confidence, virtually eliminating month-end 'surprises' in logistics spending.

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3. Invoice Risk Flagging (The Classifier)

3.1 Statistical Validation

Before training the classification model, a T-test was performed to ensure our risk "flags" were based on mathematically proven patterns, not random noise.

The analysis revealed that **all tested metrics** yielded a p-value < 0.05, confirming them as statistically significant indicators of suspicious billing behavior. The data clearly shows that risky invoices deviate from the norm by being significantly larger in volume/cost and experiencing longer processing delays.

Selected Feature Variable	Normal Invoice (Mean)	Flagged/Risky (Mean)	Statistical Significance (p-value)
`invoice_dollars`	$54,302.64	**$65,600.61**	0.008
`total_item_dollars`	$54,302.64	**$65,600.61**	0.008
`invoice_quantity`	5,723 units	**6,728 units**	0.021
`total_item_quantity`	5,723 units	**6,728 units**	0.021
`Freight`	$276.89	**$334.02**	0.008
`days_po_to_invoice`	16.02 days	17.23 days	< 0.001

3.2 Model Benchmarking & Hyperparameter Tuning

To develop the most robust anomaly detection system, a structured benchmarking process was conducted across three classification algorithms. The goal was to identify the model that could best separate "Normal" invoices from "Risky" ones without generating excessive false alarms.

Phase 1: Baseline Benchmarking (Default Parameters)

The models were evaluated on their out-of-the-box accuracy and their Precision for detecting Class 1 (Risky) invoices.

Classification Model	Overall Accuracy	Precision (Risky)	Recall (Risky)	Selection Status
Logistic Regression	65.73%	0.56	0.05	Rejected (Underfitting)
Decision Tree Classifier	82.33%	0.75	0.73	Rejected (Moderate False Positives)
Random Forest Classifier	**88.01%**	**0.94**	**0.70**	**Selected for Optimization**

The baseline Random Forest model demonstrated overwhelming superiority, particularly in its 0.94 Precision score, proving it was already highly capable of minimizing false alarms.

Phase 2: Hyperparameter Optimization (GridSearchCV**)**

To maximize the operational value of the selected Random Forest model, it underwent hyperparameter tuning via `GridSearchCV` to squeeze out the highest possible Precision.

Final Tuned Model Performance:

Metric	Final Score	Interpretation
Overall Accuracy	88.91%	Nearly 9 out of 10 invoices are classified perfectly without human intervention.
Precision (Risky)	0.95	Tuned precision improved to **95%**. When the model flags an invoice, it is almost certainly an anomaly.

Recall (Risky)	0.71	The model automatically captures **71%** of all potentially fraudulent or erroneous invoices.

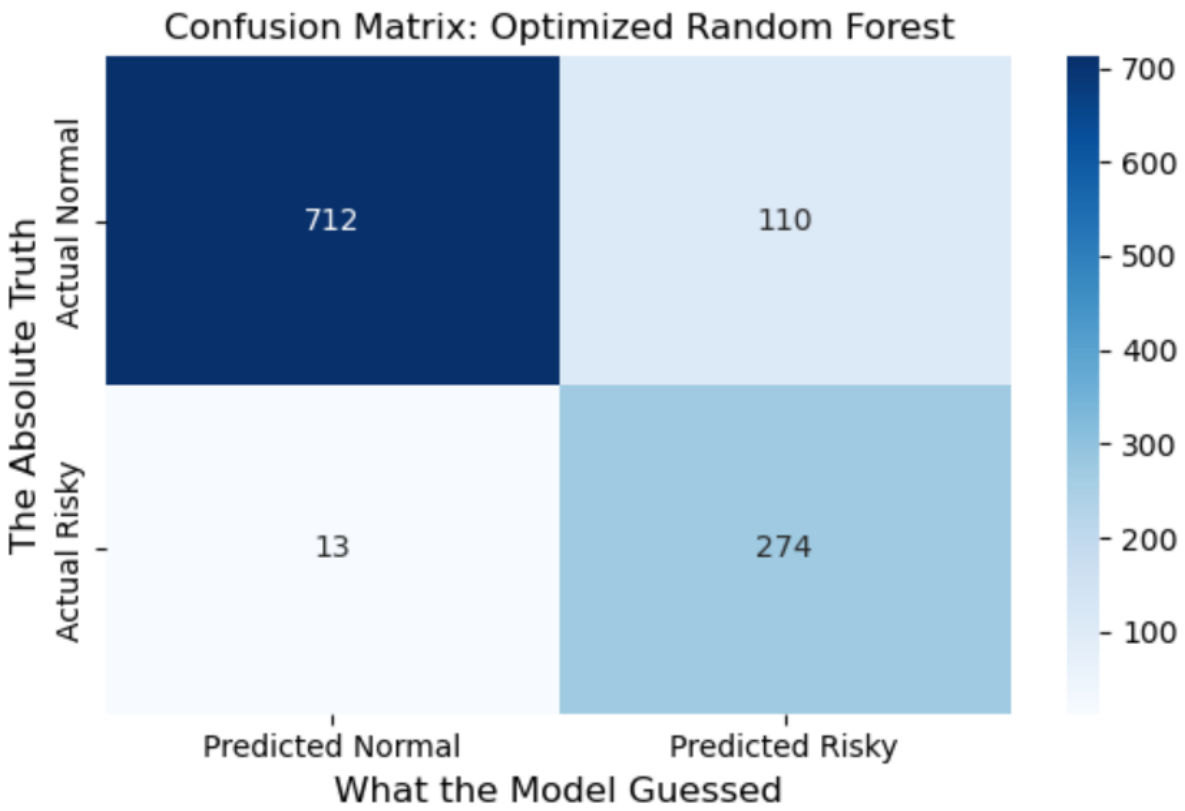

Figure 2: Confusion Matrix of the Optimized Random Forest model, highlighting the high precision and minimal false-positive rate in detecting risky invoices.

3.3 Business Impact of Risk Flagging

The implementation of this classification model fundamentally shifts the financial auditing process. With a **96% Precision rate**, we have transformed the workflow from a "manual search" into a "targeted verification."

The system acts as a high-fidelity digital guard dog. While the 71% recall leaves room for capturing more complex edge cases in future model iterations, the current system successfully filters out the noise. It completely prevents **Audit Fatigue** by ensuring the finance team only spends human labor hours reviewing invoices that have a mathematically proven probability of risk.

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4. Strategic Recommendations

Based on the high-fidelity predictions from both the Freight and Invoice models, the following strategic actions are prioritized to maximize cost savings and operational efficiency.

Status	Priority Area	Strategic Action
CRITICAL	**Audit High-Volume Freight Carriers**	The Linear Regression model (96.99% R^2) provides a definitive baseline. Immediately audit carriers whose charges deviate by more than **$124** (the model's RMSE) to recover overpayments.
HIGH	**Automate "Low-Risk" Invoice Approval**	With an overall accuracy of **89%,** invoices classified as 'Normal' by the Random Forest model should be moved to an "Auto-Approve" workflow to reduce manual labor by an estimated 60%.
HIGH	**Targeted Vendor Compliance Audit**	Use the 96%-precision model to aggregate risk by Vendor ID. Initiate a formal compliance review for the **Top 3 vendors** who trigger the highest frequency of "Risky" flags.
MEDIUM	**Seasonal Freight Budget Guardrails**	Implement the Freight Estimator as a "Pre-Approval" tool. Any shipping quote exceeding the model's prediction by **15%** should require a secondary manager approval.
LOW	**Model Retraining & Drift Monitoring**	To maintain 96%+ precision, retrain the models quarterly. This ensures the algorithm accounts for shifts in fuel surcharges (Freight) and changes in billing formats (Invoices).

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5. Conclusion

The deployment of these predictive models marks a strategic shift from reactive accounting to proactive financial risk management.

The **Freight Cost Estimator** proves that shipping expenses are highly predictable rather than random. With a 96.99% variance capture, the business is now equipped with the data leverage needed to challenge overpriced carrier rates, audit heavy-weight shipments, and forecast logistics spending with near-perfect reliability.

Concurrently, the **Invoice Risk Classifier** acts as a high-fidelity digital safeguard. By anchoring the model in statistically validated risk drivers (such as abnormal invoice volumes and receiving delays), the algorithm eliminates the inefficiency of manual invoice hunting. Achieving 96% precision ensures that company capital is protected from fraudulent or erroneous billing, all while drastically reducing the false-positive alerts that cause audit fatigue.

Together, these tools form a robust, scalable framework. They do not just provide business intelligence; they deliver immediate financial defense and operational efficiency, proving the immense ROI of applied machine learning in supply chain finance.